

Rice yield prediction for South India

Data, preprocessing and model workflow

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This guide explains where our data comes from, how we combine it, what enters the preprocessing model, what training will use, and how the application will turn user inputs into a yield estimate.

Verified dataset	Current scope
3 source families	Government crop statistics + GADM boundaries + NASA POWER weather
2,018 observations / 30 columns	Rice, four states, crop years 2001-2019
Prediction target	YIELD, measured in tonnes per hectare (t/ha)
Execution status	Master and six training CSVs generated and verified. DAE fitted for 180 epochs. Supervised yield models have definitions but are not trained. Beta UI uses illustrative outputs.

Models we will use

Model / method	Role and learning objective
SimpleImputer (median), StandardScaler, OneHotEncoder	Conventional preprocessing; training-set medians, means/scales and category mappings.
Denosing Autoencoder (DAE)	Implemented DL preprocessing stage. Reconstruct 21 clean predictors from lightly corrupted inputs; learn 8 latent features.
Gradient-boosted decision trees (GBDT)	Planned tabular baseline using HistGradientBoostingRegressor; learn nonlinear predictor-to-yield relationships.
Multilayer Perceptron (MLP) regressor	Initial tabular DL predictor: latent/context inputs -> dense layers 64 and 32 -> one yield output. Defined, not trained.
Long Short-Term Memory (LSTM); One-dimensional CNN (1D CNN)	Paper-based sequence experiments after time-series tensor preparation. Learn weather patterns over the crop season. Defined, not trained.

The current data supports yield estimation. Direct disease classification, nutrient diagnosis, and image segmentation require additional labelled data.

Three raw sources, five construction tables

Source / preserved file	Contents and contribution
Government crop statistics government/crop_statistics.csv (originally APY.csv)	State, district, crop, crop year, season, reported area and production. We derive rice yield = production / area.
GADM 4.1 administrative level 2 boundaries/india_districts.geojson.zip	District boundaries, state/district names and identifiers. We match names and calculate representative coordinates.
NASA POWER Daily API extract weather/daily_weather.csv	Daily temperature, relative humidity, wind, precipitation, radiation, surface pressure and evapotranspiration proxy, queried at matched district locations. Vapour pressure is calculated from temperature and humidity.

These preserved files live under **DATASET/raw/**. Their contents match the original project copies. The weather CSV is an assembled API extract. It is not a second government label dataset. Preserve raw files without editing.

Temporary join tables - no extra user-facing datasets

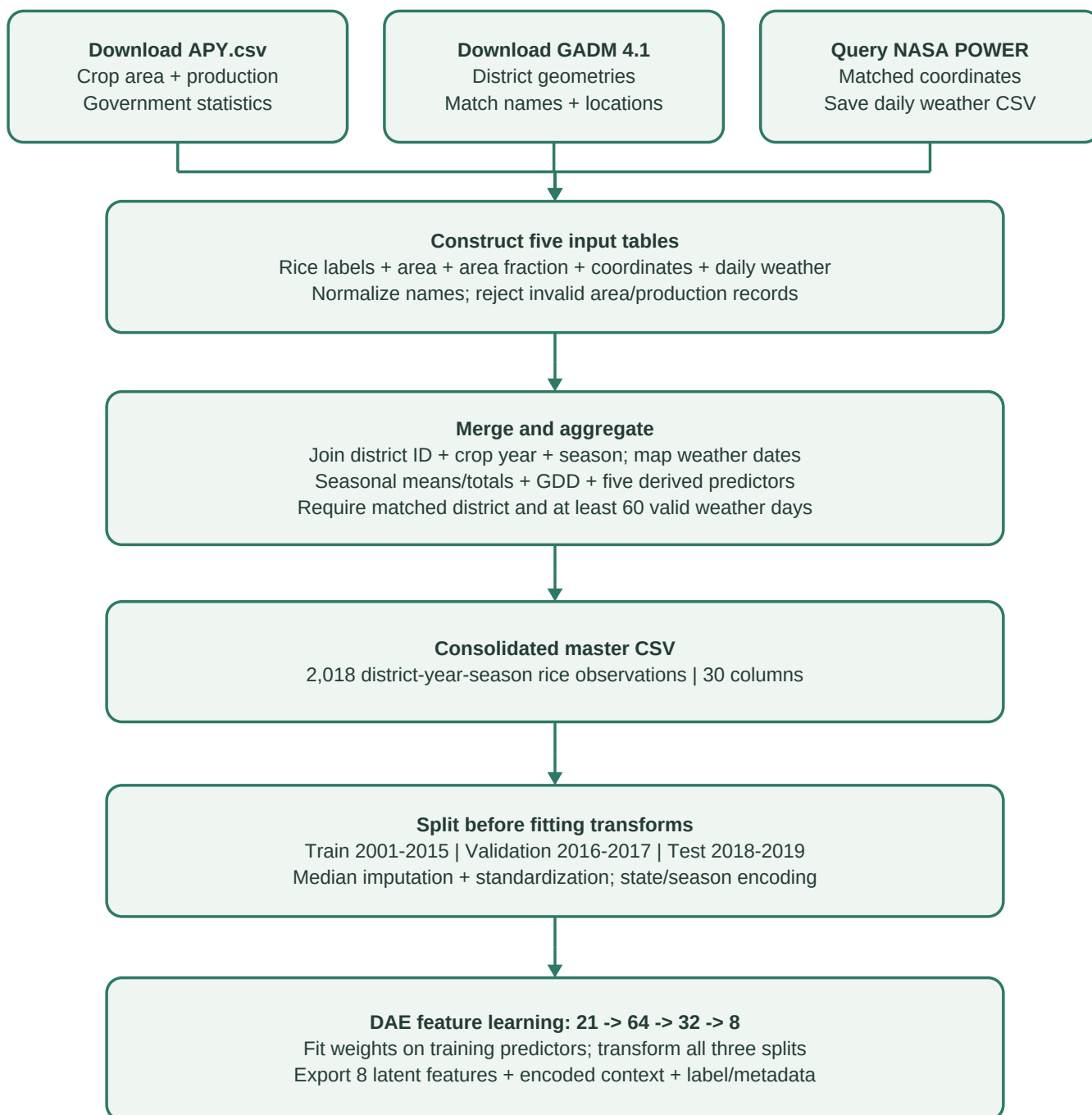
File	Contents
YIELD_NUTS2_SOUTH_INDIA_RICE.csv	3,554 rice labels; yield, production, year, season and district ID.
CROP_AREA_NUTS2_SOUTH_INDIA_RICE.csv	Rice cultivated area for each district-year-season.
AREA_FRACTIONS_NUTS2_SOUTH_INDIA_RICE.csv	District rice area / recorded state rice area, for the same year and season.
CENTROIDS_NUTS2_SOUTH_INDIA.csv	District crosswalk, coordinates and match status.
METEO_DAILY_NUTS2_SOUTH_INDIA.csv	Daily weather used for season-specific aggregation.

The raw crop selection yields 66,172 valid four-state records across crops; our final experiment selects rice. The builder creates the five join products temporarily, then removes temporary files automatically. The only retained merged CSV is **DATASET/merged/rice_master.csv**. Netherlands data contributes no rows. FAPAR, soil, elevation and WOFOST variables are not currently included.

Source links

[Government crop-statistics resource](#) | [GADM data](#) | [NASA POWER Daily API](#)

From downloads to preprocessed training files



Verified outputs: DATASET/training/standard/{train,validation,test}.csv and DATASET/training/autoencoder/{train,validation,test}.csv. Models live in pre-processing/artifacts/; plots/reports live in docs/. Raw sources are preserved separately.

What the 30 master columns represent

The builder joins yield, area and area fraction using **IDREGION + FYEAR + SEASON**. It joins district coordinates by IDREGION and selects weather by district and date. Each output row describes one rice observation, not one day or one farm.

Group	Exact column names and meaning
Identity / context	CROP; IDREGION; STATE; DISTRICT; FYEAR; SEASON. Observation identity, location and crop-calendar context.
Label / reported production	YIELD (t/ha); PRODUCTION_TONNES (tonnes). Yield is the prediction label; production is provenance only.
Crop area	CROP_AREA (ha); AREA_FRACTION (district share of state-season rice area); CROP_AREA_LOG1P (log(1 + area)).
Location / matching	CENTROID_X (longitude); CENTROID_Y (latitude); MATCH_STATUS (district-name crosswalk status).
Weather coverage	WEATHER_DAY_COUNT: number of valid daily records used.
Temperature	TMAX_MEAN, TMIN_MEAN, TAVG_MEAN (degrees C); TEMP_RANGE_MEAN = max mean - min mean; GDD_BASE10 = sum(max(TAVG - 10, 0)).
Atmospheric conditions	VPRES_MEAN (hPa), WSPD_MEAN (m/s), RELH_MEAN (%), PS_MEAN (kPa).
Season totals	PREC_TOTAL (mm), ET0_TOTAL (mm land-evaporation proxy), RAD_TOTAL (sum of daily radiation, kWh/m2 for the original RE-community API query).
Derived interactions	WATER_BALANCE_PROXY = PREC_TOTAL - ET0_TOTAL; RAIN_PER_GDD = rainfall / GDD; RADIATION_PER_GDD = radiation / GDD. Zero GDD uses zero for ratios in the current builder.

Weather season assumptions

Kharif: June-November. Rabi: preceding November-April. Summer: March-June. Autumn: September-December. Winter: preceding December-March. These are approximate month windows, not district-specific recorded sowing dates.

The builder calculates source-derived features and retains actual label observations. Calling this dataset synthetic would misdescribe how it was made. ET0 uses NASA EVLAND land evaporation as a proxy; it is not a calibrated FAO-56 reference ET estimate. API unit metadata was checked: the RE-community radiation values are kWh/m2/day, so their seasonal sum is kWh/m2.

What changes, what is excluded, what is learned

Stage	Input -> processing -> output
Split	2,018 rows -> chronological partition -> 1,485 train / 245 validation / 288 test.
Median imputation	21 numeric columns -> fit medians using training rows -> fill missing values consistently in every split.
StandardScaler	Imputed numeric values -> fit training mean and standard deviation -> comparable z-score inputs.
OneHotEncoder	STATE and SEASON -> fit training categories -> categorical indicator columns. Unseen categories are ignored by the current encoder.
DAE training	21 standardized predictors + small Gaussian corruption -> reconstruction objective -> 8 learned latent features. Clean inputs are the reconstruction targets; yield is not used.
DAE export	Apply fitted encoder without training noise -> AE_LATENT_01 to AE_LATENT_08; reconstruction MSE; encoded categories; retained label and identifiers.

The 21 numeric inputs

CROP_AREA, AREA_FRACTION, CENTROID_X, CENTROID_Y, WEATHER_DAY_COUNT, TMAX_MEAN, TMIN_MEAN, TAVG_MEAN, VPRES_MEAN, WSPD_MEAN, PREC_TOTAL, ET0_TOTAL, RAD_TOTAL, RELH_MEAN, PS_MEAN, GDD_BASE10, TEMP_RANGE_MEAN, WATER_BALANCE_PROXY, RAIN_PER_GDD, RADIATION_PER_GDD, CROP_AREA_LOG1P.

Columns excluded from model inputs

Column(s)	Reason / treatment
PRODUCTION_TONNES	Excluded to prevent leakage: production/area defines the yield target.
YIELD	Kept as the supervised label, excluded from preprocessing-model inputs.
CROP	Constant rice crop in this cohort; retained only in the master.
IDREGION, DISTRICT	Identifiers; retain for tracing predictions, exclude from encoder inputs.
FYEAR	Defines chronological splits; not an encoder input in this implementation.
MATCH_STATUS	Provenance only; not a predictor.

No master columns are physically deleted. The deep CSV replaces the 21 explicit numeric predictors with 8 latent columns and retains selected metadata. Both current representations use the same 21 original numeric predictors and the same observation splits. Conventional exports have 28 predictor columns; DAE exports have 16 including reconstruction error.

What preprocessing has taught us so far

Measured from the current master	Finding
Cohort filtering	3,554 rice labels -> 2,018 usable master rows. 1,536 label rows do not enter the final matched/covered cohort; the current builder does not export a per-reason rejection audit.
State coverage	Andhra Pradesh: 572; Karnataka: 722; Tamil Nadu: 612; Telangana: 112. Coverage is unequal.
Season coverage	Kharif: 1,037; Rabi: 492; Summer: 345; Winter: 75; Autumn: 69.
Time coverage	2001-2019. Training: 2001-2015; validation: 2016-2017; test: 2018-2019.

What the generated preprocessing plots show

Coverage charts show imbalance. Yield distributions and annual curves reveal unusual values and temporal patterns. Correlation charts show associations and redundant features. Raw-versus-standardized histograms demonstrate the effect of scaling, not improved accuracy.

What the DAE learns and how we assess it

The encoder learns numerical dependencies among weather, location and crop predictors. Training/validation reconstruction losses assess how well it reproduces the clean standardized inputs.

Reconstructed-versus-original plots show where it loses information. High reconstruction error means the representation fits that row poorly; it does not prove a source error or diagnose disease.

The eight latent dimensions have learned numerical meaning. We cannot claim that one is drought, another is nutrient deficiency, or that lower reconstruction loss guarantees better yield estimates.

Measured DAE run and remaining evaluation

The DAE completed 180 epochs. Validation chose epoch 178 with reconstruction MSE **0.07724** in standardized feature units. All exported predictors are finite, labels and chronological splits are preserved, and the master has no missing cells or duplicate district-year-season keys. Supervised yield accuracy and superiority over the reference paper have not been established.

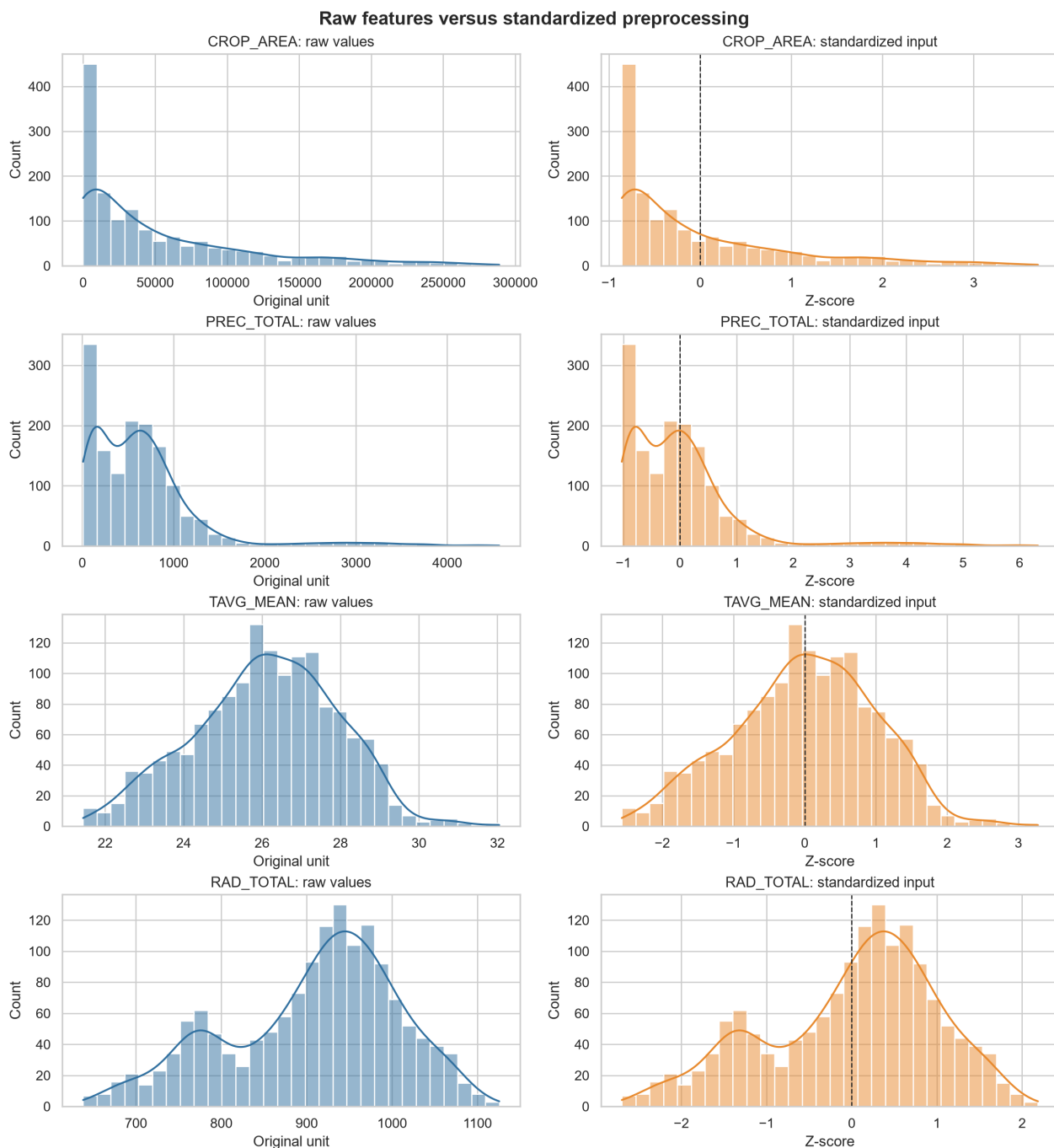
Autumn and Winter are absent from training years. Their season categories map to zero under the current encoder rule. Evaluate those seasons separately; a pooled score alone can hide this generalization limit.

Compare identical test rows using MAE (t/ha), RMSE (t/ha), R2, and state/season errors. Use validation for selecting model settings. Our Indian rice results cannot directly be ranked against European crops tested on different datasets.

Important data assumptions

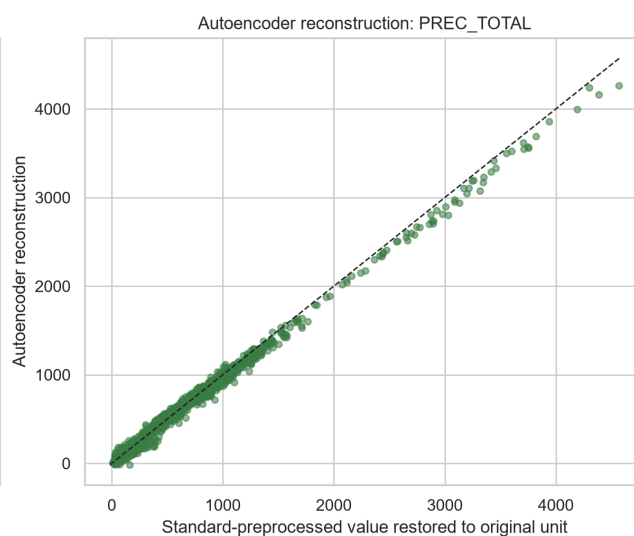
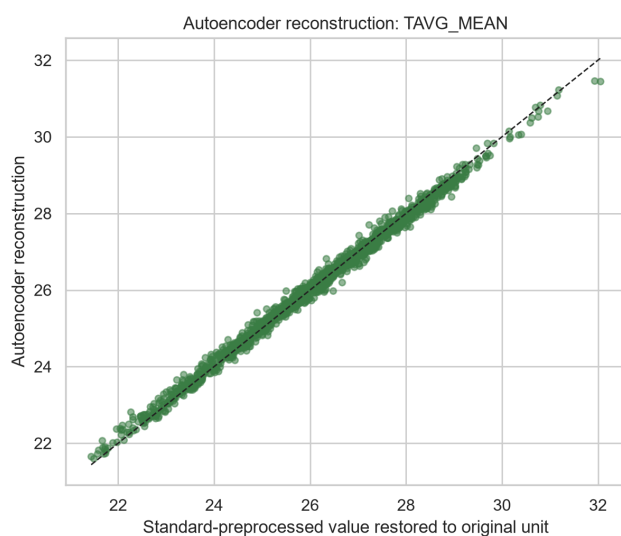
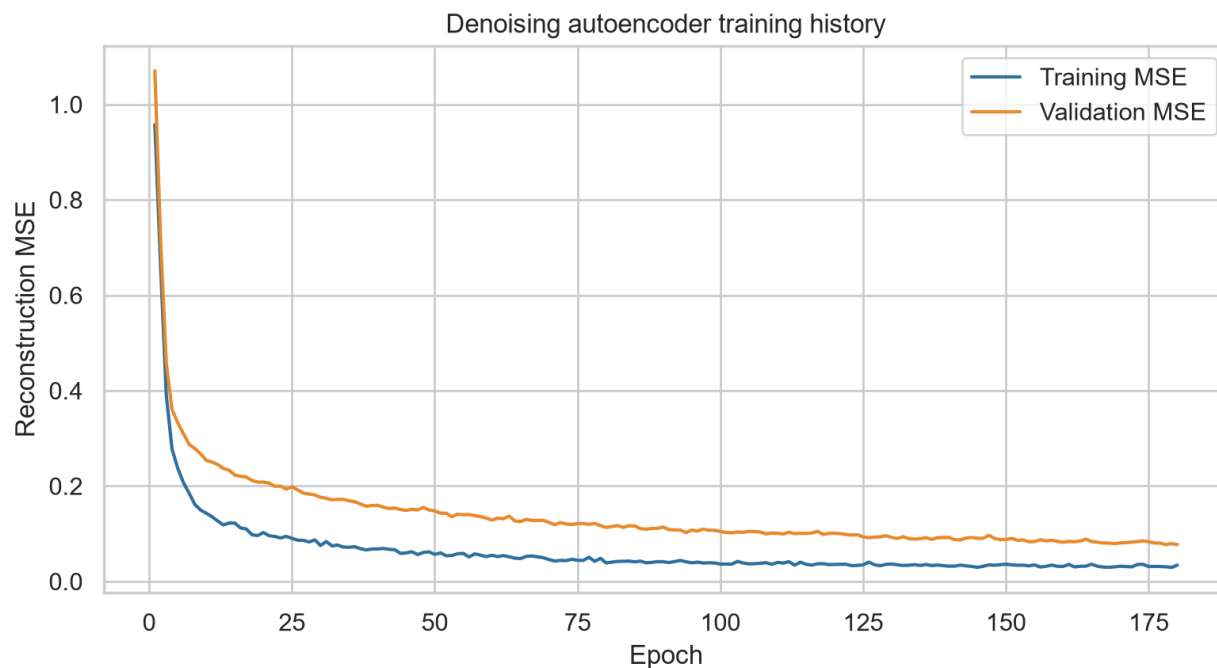
District weather is reanalysis, not farm sensor data. Coordinates currently average boundary vertices. Partial-season coverage can bias totals despite the 60-day rule. Full-season weather and reported area support retrospective estimation; early forecasts need predictors restricted to the forecast date.

Raw values and standardized model inputs



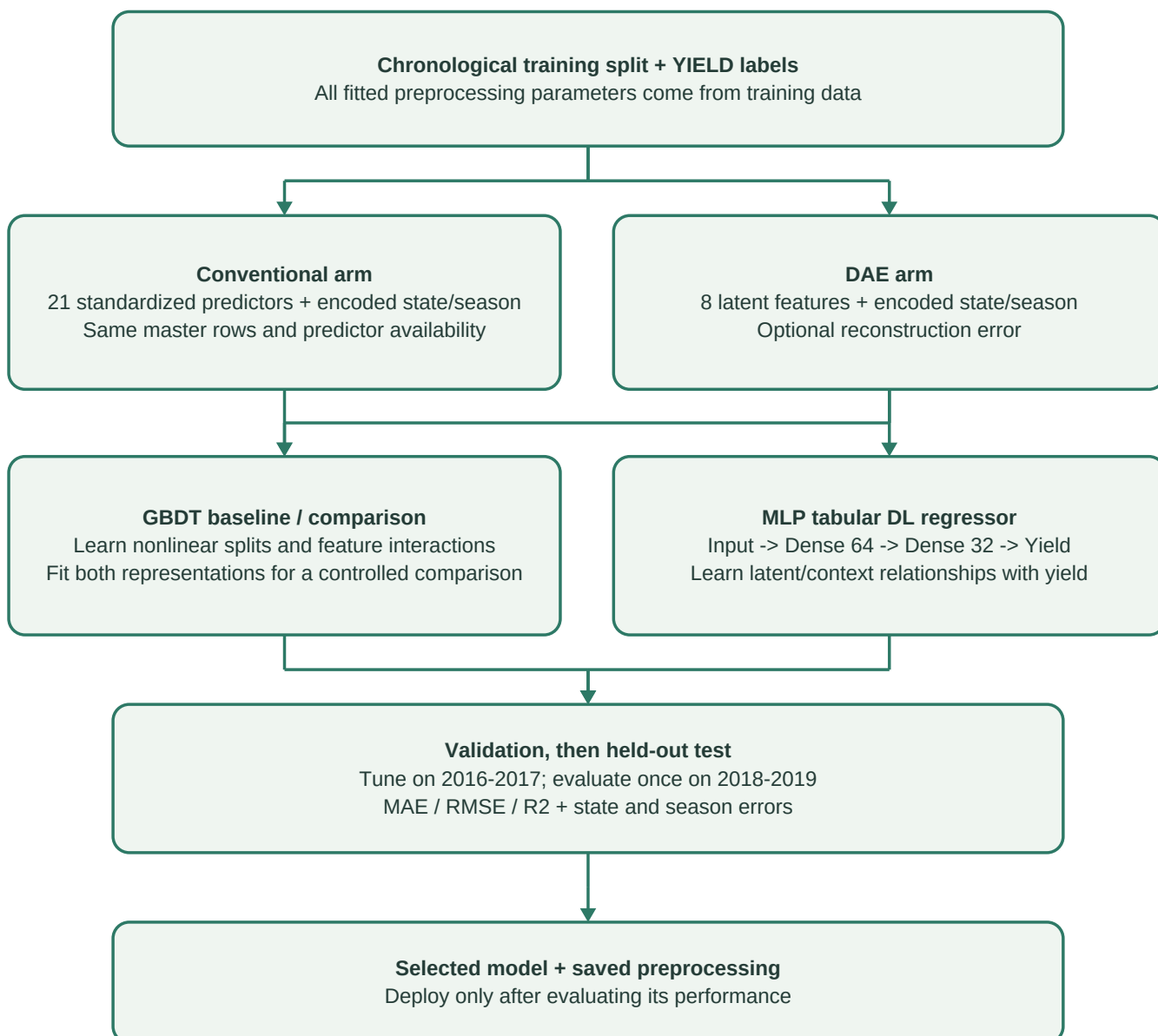
These are actual training observations before and after StandardScaler. Differences in shape or scale illustrate transformations; they are not proof of higher prediction accuracy. Imputation/scaling parameters come only from training years.

DAE learning and reconstruction



Actual 180-epoch run; best validation checkpoint: epoch 178. The dashed diagonal indicates perfect reconstruction for the selected raw-unit features. These plots evaluate feature reconstruction. Yield accuracy still requires supervised-model training and held-out tests.

How preprocessed data becomes a yield model

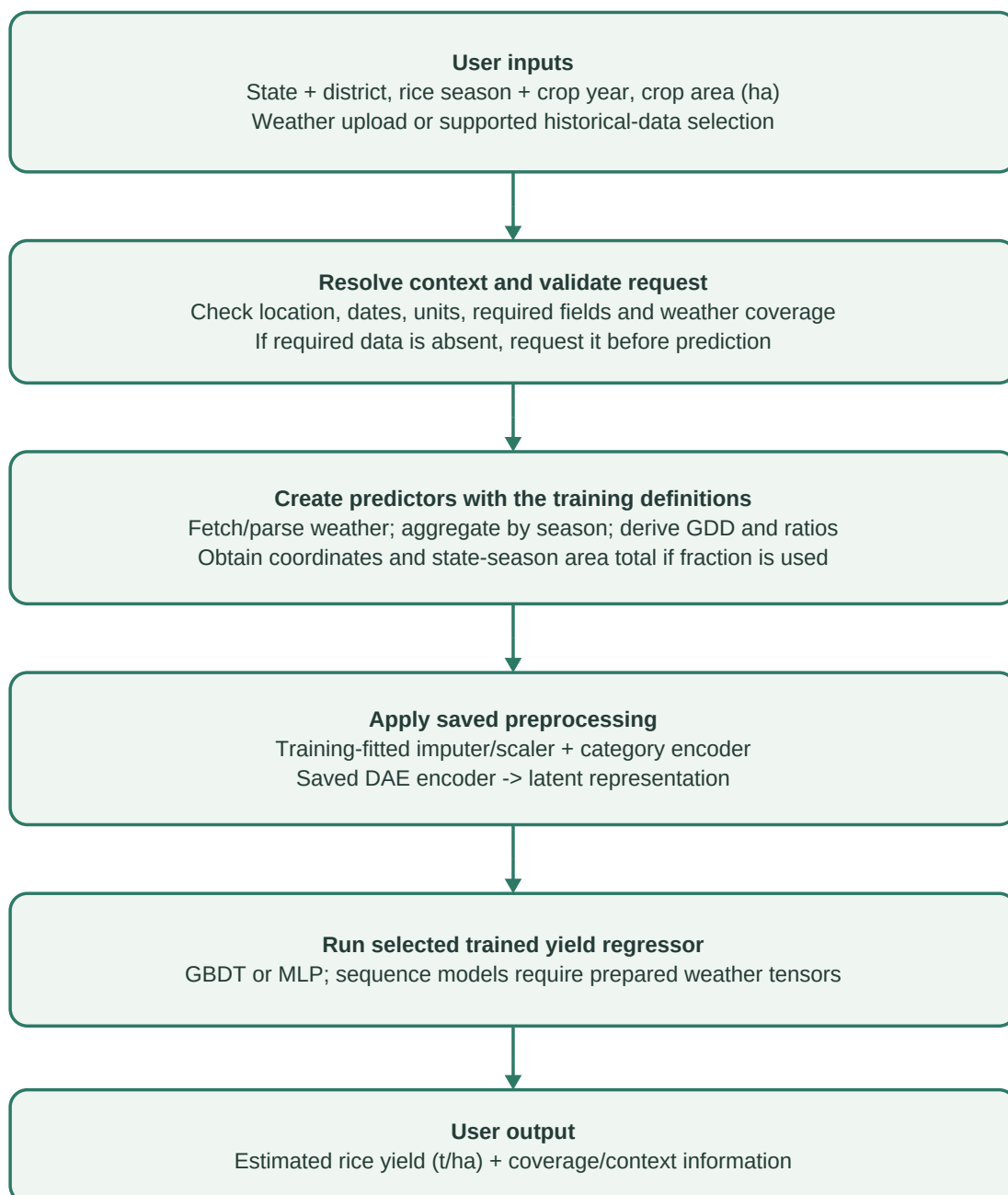


Paper-based sequence experiment

Prepare district-season sequences from daily/dekad weather using the same observation split. **LSTM** learns patterns across successive weather steps; **1D CNN** learns local time-window patterns with temporal filters. Combine their sequence representation with static/context inputs and a regression head. These sequence tensors and forecasting trainers are planned work; the master CSV alone is not a weather sequence.

The DAE is an adapted implementation of the established Vincent et al. preprocessing method. The LSTM/1D CNN experiments follow the model families in Paudel et al. (2023); this is an Indian-data adaptation, not an exact replication of its European multimodal experiment.

What the user enters and what they receive



The user does not enter a known yield or observed production. The service calculates derived inputs; district names identify the request but are not numerical encoder features. AREA_FRACTION requires a corresponding state-season rice-area denominator: obtain that context or redesign/retrain the feature set.

Optional output: estimated production = predicted yield x user crop area. This is derived from the prediction, not an input. Confidence/risk bands need a separately validated uncertainty method. Current data cannot produce a crop disease label.

Where everything lives and what remains to build

Project navigation

Folder	Purpose
APP/	Implementation and existing application scaffold.
DATASET/raw/	Three preserved source files; clear government/boundaries/weather names.
DATASET/merged/	Only rice_master.csv and its README; 2,018 rows x 30 columns.
DATASET/training/	standard/ and autoencoder/ split CSVs plus predictor manifest.
pre-processing/	Dataset builders and preprocessing scripts.
docs/	Current guide, reports and reference papers; source/ contains this PDF generator.
github/	GitHub working copy and repository-related files.
OTHER/archive/	Superseded material, recoverable and excluded from training/GitHub.

Files consumed by supervised training

Conventional: DATASET/training/standard/train.csv, validation.csv, test.csv. Deep: DATASET/training/autoencoder/train.csv, validation.csv, test.csv. Select predictor columns explicitly using DATASET/training/manifest.json and separate YIELD. Only train.csv fits supervised weights; validation selects settings; test measures final performance.

Next implementation steps

Preprocessing is executed and verified. APP/models/ contains GBDT, MLP, LSTM and 1D CNN definitions. Next: implement supervised trainers and sequence preparation; evaluate held-out years; save the selected predictor; connect it to the beta UI. The prototype is available at APP/prototype/index.html and uses fixed, explicitly illustrative yield outputs.

Hyperlinked references

1. Paudel et al. (2023), Interpretability of deep learning models for crop yield forecasting.
2. Vincent et al. (2008), Extracting and Composing Robust Features with Denoising Autoencoders.
3. scikit-learn HistGradientBoostingRegressor documentation.
4. Project dataset repository.
5. European reference dataset record - background only.

A simple entry point for the team

Read docs/README.md, then use these commands from the project or GitHub clone root:

```
python -m pip install -r pre-processing/requirements.txt
python pre-processing/run_pipeline.py --epochs 180
python pre-processing/verify_outputs.py
python -m http.server 8765 --directory APP/prototype
```

Open <http://localhost:8765> for Verdant, the green/off-white beta. Overview shows source and dataset counts; Yield preview demonstrates user input and result cards; Data pipeline explains files; Models & learning names every method and its role.

The beta is a frontend demonstration. Its displayed 4.20 t/ha is fixed and explicitly labelled illustrative; multiplying it by entered area produces illustrative tonnes. No trained yield model runs in the prototype.

Checklist

Requested item	Delivered / status
Sources, counts, contents, merge and column exclusions	Documented here, in source code and dataset READMEs.
Full preprocessing model and trainable outputs	DAE fitted; standard and DAE CSVs verified; checkpoint and transforms saved.
Three end-to-end diagrams	Data construction; training architecture; user computation flow.
Exact computation model names	HistGradientBoostingRegressor (GBDT), MLP, LSTM and 1D CNN; supervised weights are not yet trained.
Clean local and GitHub structure	Active data has raw, merged and training stages. Historical local material is archived outside the workflow.
Prototype without yield training	Verdant beta: navigation, dependent district selectors, form validation and illustrative result cards.

The DAE is preprocessing; it does not itself predict crop yield. The selected supervised yield predictor will provide the application estimate after training and evaluation. LSTM/1D CNN require time-series inputs; MLP/GBDT use the prepared tabular representations.

The proposed application experience

Workspace / Yield preview

Interface prototype - v0.1

RICE YIELD STUDIO

Overview

Yield preview

Data pipeline

Models & learning

HARVEST OUTLOOK / DEMONSTRATION

Make the season come into focus.

Try the proposed input-to-output experience. No trained yield model runs in this preview.

Demo mode The displayed 4.20 t/ha is a fixed illustrative value, independent of weather or model inference.

Your crop context

State: Karnataka

District: Mandya

Season: Kharif

Crop year: 2019

Crop area (hectares): 20

Crop: Rice

Seasonal weather context

Future service: NASA POWER lookup or daily-weather upload. This prototype uses no weather API.

Refresh harvest preview

Coordinates, GEO, seasonal summaries and area fraction will be calculated by the future service.

YOUR HARVEST PREVIEW

Illustrative yield - not a trained prediction

4.20 t/ha

Mandya - Kharif 2019 - 20 ha

Illustrative production

84 tonnes

Demo yield x your crop area

No confidence interval or disease diagnosis is calculated. Model performance will be added after validation.

Inside the future computation

1. Validate location, season and area
2. Obtain weather and create 21 predictors
3. Apply fitted input, scaler and encoding
4. Encode using the Denoising Autoencoder
5. Run a trained MLP or GBDT regressor
6. Return rice yield in tonnes/hectare

These stages are illustrated here: only input validation and the fixed demo calculation run.

Verdant - South India rice research

Interface preview. Every estimate is explicitly illustrative.

Four-state research cohort

Built for a clearer view of the growing season.

B Bhat - Joshith G

The verified demonstration selected Karnataka / Mandya, Kharif 2019, and 20 hectares. The result card shows fixed illustrative yield 4.20 t/ha and illustrative production 84 tonnes. This checks interface calculations only.

The sidebar includes Overview, Yield preview, Data pipeline and Models & learning. The model screen explicitly names the DAE preprocessing network, conventional transforms, GBDT baseline, MLP regressor and future LSTM/1D CNN sequence models.

Open APP/prototype/index.html directly or serve it with the command on the previous page. No supervised yield weights or live weather service are connected to the beta.